

Arnav Patil

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EDUCATION

University of Toronto

Sept 2023 – Apr 2028

BASc. in Computer Engineering with PEY Co-op

Toronto, ON

- **GPA: 3.63/4.00** with recognition on Dean's Honours List
- **Minor in Engineering Business**

EXPERIENCE

Digital IC Design Intern

May 2026 – Present

Marvell Technology

Toronto, ON

- Register-transfer level logic for PHY-layer die-to-die connectivity IP in TSMC 2nm and 3nm FET technologies
- Debugging RTL simulation workflows and testbenches in VCS to support chip bring-up & validation across teams

Teaching Assistant

Sept 2025 – Dec 2025

University of Toronto

Toronto, ON

- Facilitated tutorials for **APS100 Orientation to Engineering**, reinforcing concepts through discussion
- Provided mentorship on academic strategies, course planning, and career development to first-year students

FPGA Engineer Intern

May 2025 – Aug 2025

Intelligent Sensory Microsystems Lab

Toronto, ON

- Developed FPGA-based modules for novel CMOS time-of-flight sensors in Prof. Roman Genov's **ISML** group
- Achieved 800 Mbps sensor readout throughput by designing high-speed RTL modules on Artix-7 FPGAs
- Implemented SPI and I2C controllers in Verilog to handle sensor initialization, enabling high-speed readout

PROJECTS

MOSAIC: Multi-Objective Smooth Analytical Placer |

- Submission to the **2026 Macro Placement Challenge** co-hosted by Partcl and Hudson River Trading
- Achieved a 10.3% improvement on RePlAce on IBM ICCAD04 benchmarks, averaging a 99 s/bench runtime
- Implemented differentiable approximations for the TILOS MacroPlacement evaluator to minimize cost in Python

SpecNet: Deep Learning for Analog Circuit Design |

- Designed a neural network pipeline to automate transistor-level analog circuit sizing from target specifications
- Achieved 98% accuracy within 5% of optimal dimensions, greatly reducing number of SPICE simulations required
- Automated dataset generation by parsing LTspice simulation logs into structured ML training data using Python

DE1-SoC Arcade Game |

- Implemented a blackjack game in Verilog on an FPGA, managing FSMs for game logic and player interaction
- Simplified hardware peripheral control from top-level modules by writing API-like wrapper RTL over Altera IP
- Fixed version compatibility issues in provided IP by writing a Python script to re-format memory files (.mif)

Nios-V Ultrasound System |

- Integrated an ultrasonic sensor and servo motor into the FPGA-based Nios-V soft processor via GPIO ports
- Designed & implemented BJT pull-up/pull-down networks for safe voltage conversion between FPGA and sensor
- Implemented timer-based polling in C to calculate object distance from 20 cm to 2 m within 2 cm precision

Spatial Audio Engine on ARM Cortex-M4 |

- Built a real-time DSP pipeline on a Cortex-M4 MCU, processing stereo audio streams end-to-end in bare-metal C
- Implemented fixed-point arithmetic to eliminate floating-point overhead, maximizing computational throughput
- Engineered dynamic stereo audio by computing channel gain and phase delay to perceive an accurate position

COURSEWORK

Languages & Tools: C/C++, (System)Verilog, Python, RISC-V Assembly, Quartus Prime, Vivado, Virtuoso

Hardware Courses: Computer Architecture, Digital & Analog Electronics, Computer Hardware, Signal Processing

Software Courses: Operating Systems, Embedded Systems & Firmware Programming, Machine Learning